

# Snehin Adepu

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Hyderabad, India

snehin-adepu



Software Engineering student with hands-on experience building backend and ML-powered systems in production-like environments. Reduced LLM inference costs by 70%+, deployed services on GCP with CI/CD and Docker, and developed full-stack apps using FastAPI, React, and PostgreSQL. Strong foundation in data structures, algorithms, OS, and DBMS, with a focus on high-performance APIs and scalable infrastructure.

## EDUCATION

**B.TECH, COMPUTER SCIENCE (AIML) | CGPA: 8.55 | 2023 - PRESENT**

VNR VIGNAN JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY, HYDERABAD

**INTERMEDIATE | GRADUATED WITH 95.3% | JUNE 2021 — MAY 2023**

SRI CHAITANYA JUNIOR COLLEGE, HYDERABAD

**10TH GRADE | SECURED 89% | JUNE 2020 — MAY 2021**

SURYA THE GLOBAL SCHOOL, HYDERABAD

## TECHNICAL SKILLS

**Languages:** Java, Python, C, JavaScript

**Web & Backend:** FastAPI, Node.js, Express, React, Vite, TailwindCSS

**Databases:** PostgreSQL (Supabase), MySQL

**Cloud & DevOps:** Google Cloud Platform, AWS, Docker, Terraform, Git, GitHub, CI/CD

**ML/AI:** CNNs, XGBoost, Prompt Engineering, Model Benchmarking

**Tools:** Postman, Artillery, k6, Figma, Tableau

**Core Concepts:** Data Structures & Algorithms, OOP, REST APIs, System Design Basics

## EXPERIENCE

**AIML & Backend Intern — Loomylabs | May 2025 - Nov 2025**

- Designed early-stage backend architecture by evaluating frameworks, cloud services, and AI tooling.
- Benchmarked multiple LLMs and developed optimized prompting pipelines improving output reliability and reducing inference cost by 70%+.
- Managed complete GCP deployment including VM provisioning, networking, CI/CD setup, and DNS management.
- Designed and optimized Supabase Postgres schemas and queries for faster data access and efficient workflows.
- Implemented multi-layer AI cost-optimization (smart routing, caching, summarization, context limiting, and RAG tuning).
- Performed high-scale load testing with Artillery and k6, ensuring stability with 1000+ concurrent users.
- Contributed to SEO strategy and technical assessments for backend/AI candidates.

## ACADEMIC PROJECTS

### Reliable SMTP Email System

- Built a full-stack SMTP email pipeline (Node.js, Express, Nodemailer, Supabase) with retry logic and delivery tracking.
- Developed a real-time React dashboard with email status polling and error metrics visualization.

### Smart Irrigation System (IoT)

- Created an IoT-based irrigation controller using soil-moisture sensors to reduce water usage.
- Added real-time monitoring and automated control logic for optimized irrigation.

### Expense Tracker (Java + AWT + JDBC)

- Built a personal finance manager with expense categorization and dynamic MySQL-based reporting.

### Historical Artifact Management System (SQL)

- Designed a 3NF relational database with ER diagrams, keys, indexes, and stored procedures for fast retrieval.

### Uber Rides Analysis (ML)

- Analyzed Uber ride data and built fare-prediction models using feature-driven ML techniques.

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## ACHIEVEMENTS

### Runner-Up - Startup Race

- Developed and pitched a groundbreaking idea focused on smart electricity management and strategic deployment of EV chargers.
- Outperformed several teams in a competitive environment to achieve the second position.

### Sponsorship Committee Member, Inter-College MUN

- Contributed to organizing an inter-college MUN event attended by 300+ delegates.
- Played a key role in sponsor outreach, event planning, hospitality coordination, and ensuring smooth execution.

### Runner-Up - Blind UI/UX Challenge, Synergia 2025 (BVRIT Hyderabad)

Secured 2nd place in a rapid-memory UI/UX replication challenge; recognized for speed, accuracy, and design precision.

## EVENTS & PARTICIPATION

### KRITHOATHON Hackathon - Participant

- Collaborated on a project focused on smart irrigation leveraging moisture sensors.
- Designed and implemented a system to optimize water usage in agriculture based on soil moisture levels.

### Open House 2K24 - Participant

- Presented the project "Transforming Floral Waste into Valuable Products" at VNRVJiet, Hyderabad.
- Proposed an innovative approach to repurpose floral waste into sustainable products.
- Received constructive feedback from a panel of experts to enhance the project's scalability and impact.

### HackWave 2025 - Round 2, SNIST

- Selected to Round 2 out of 1100+ teams, with only 80 teams advancing.
- Built Cultural Context Analyzer, a full-stack AI tool using Google Gemini for cultural & historical text analysis.
- Tech Stack: FastAPI, React + Vite, Supabase (PostgreSQL), TailwindCSS.
- Implemented multilingual support (12+ languages), cultural insights, interactive maps, and timelines.
- Received positive feedback from jury for innovation and real-world applicability.

### Synergia 2025 - BVRIT Hyderabad

- Participated in Bug Bounty and Blind UI/UX challenges; achieved 2nd place in the Blind UI/UX event and collaborated with peers across multiple technical competitions.

## LEADERSHIP & ACTIVITIES

### Design Lead - Krithomedh (AI/ML Club)

- Led branding, poster design, and event visual development for technical events.

### Documentation Volunteer - ISOC Organization VNRVJiet

- Led the documentation team, overseeing planning, content creation, and structured reporting for technical and organizational activities.

## RESEARCH & PUBLICATIONS

### Conference Paper Presentation - ICONFEST 2025

#### "Securing IoT Networks Using Hybrid Machine Learning Model Against DDoS Attacks"

- Presented a research paper at ICONFEST 2025 proposing a hybrid XGBoost-CNN IDS that achieved 99.03% DDoS detection accuracy with automated firewall-based mitigation for IoT networks.

### Conference Paper Presentation - ICICA 2026

#### "Home Garden Irrigation Control System Employing Hexagonal Layout and Geometric-Depth Moisture Sensing"

- Presented a smart home garden irrigation system using hexagonal sensor layout and geometric-depth moisture sensing, reducing water consumption by ~57% through Arduino-based automated control.